

MICROSOFT EXCEL

COMPLETE PROFESSIONAL VISUAL STUDY MANUAL

Paid Enrollee Course Material

Course Code: EDA-EXC-103

Level: Beginner to Intermediate

Duration: 6 Weeks (18 Lessons)

Certificate: Professional Certificate in Microsoft Excel

Prerequisite: Computer Appreciation

Learning Format: Classroom + Guided Practical + Independent Practice

COURSE OVERVIEW

Microsoft Excel is a spreadsheet application used to organize data, perform calculations, analyse information, create charts and prepare reports. This course develops practical spreadsheet skills for school, office, business and professional work.

The course is designed as a full teaching manual. Learners work with realistic examples such as budgets, student records, sales data, attendance, course fees and management dashboards.

LEARNING OBJECTIVES

- Understand workbooks, worksheets, rows, columns, cells and ranges.
- Enter and edit text, numbers, dates and formulas correctly.
- Format worksheets professionally.
- Use arithmetic formulas and core functions such as SUM, AVERAGE, MAX, MIN and COUNT.
- Understand relative, absolute and mixed references.
- Apply IF and other useful logical functions.
- Sort, filter and validate data.
- Use conditional formatting.
- Create appropriate charts.
- Use lookup functions to retrieve information.
- Create PivotTables for summaries.
- Prepare print-ready worksheets and dashboards.

COURSE OUTLINE

Lesson	Topic
1	Excel Foundations and Interface
2	Workbooks, Worksheets, Cells and Data Types
3	Entering, Editing and Managing Data
4	Formatting Professional Worksheets
5	Formulas and Operators
6	Core Functions: SUM, AVERAGE, MAX, MIN and COUNT
7	Cell References and Copying Formulas
8	Logical Functions and IF
9	Sorting, Filtering and Excel Tables
10	Data Validation and Conditional Formatting
11	Charts and Data Visualization
12	Text, Date and Useful Business Functions
13	Lookup Functions
14	PivotTables and PivotCharts
15	Page Layout, Printing and PDF Export
16	Data Cleaning and Error Checking
17	Business Reporting and Dashboard Design
18	Final Professional Excel Project

LESSON 1: EXCEL FOUNDATIONS AND INTERFACE

Microsoft Excel is a spreadsheet application. A spreadsheet organizes information in a grid and can calculate, analyse and present that information.

Microsoft Excel Interface

Simplified worksheet learning example

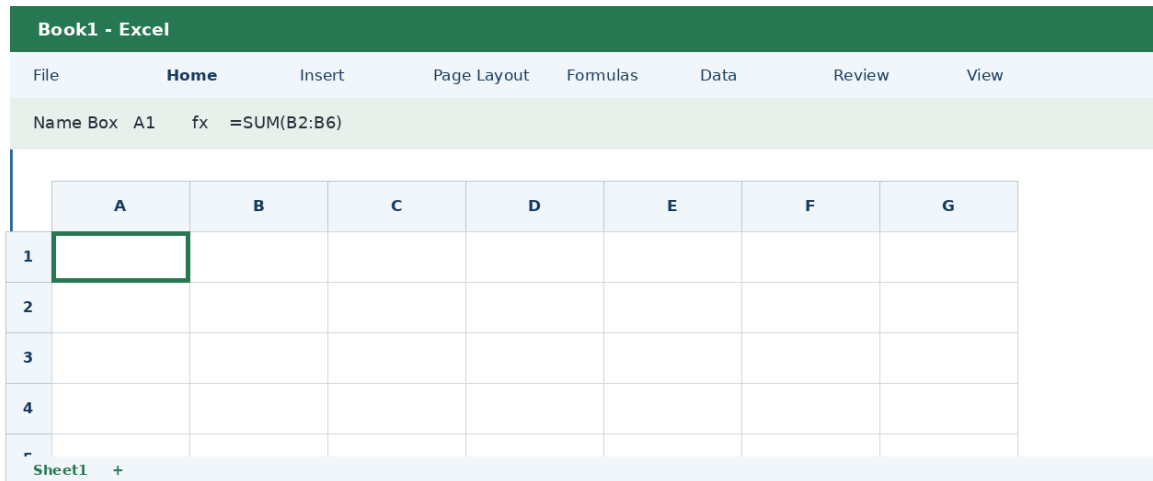


Figure 1. The main areas of a Microsoft Excel worksheet.

Workbook and Worksheet

A workbook is an Excel file. A workbook can contain one or more worksheets. Each worksheet contains rows and columns.

Key Interface Areas

- Title bar
- Ribbon and tabs
- Name Box
- Formula Bar
- Worksheet grid
- Sheet tabs
- Status bar and zoom controls

Why Excel Matters

Excel is used for budgets, payroll support, attendance, inventories, sales records, academic scores, financial analysis and management reporting.

Step-by-Step Practical

1. Open Excel and create a blank workbook.
2. Identify the Ribbon, Name Box, Formula Bar, worksheet grid and Sheet1 tab.
3. Click cell A1 and observe its address in the Name Box.

4. Save the workbook as EDA_Excel_Practice.xlsx.

Review Questions

1. What is a workbook?
2. What is a worksheet?
3. What is a cell?
4. What is the Formula Bar used for?

Assignment

Create a labelled sketch of the Excel interface.

LESSON 2: WORKBOOKS, WORKSHEETS, CELLS AND DATA TYPES

Excel stores information in cells. A cell address combines its column letter and row number, for example B4.

Cells, Data Types and References

Every value belongs in a cell address such as B4

COMMON DATA TYPES	
Text	Student Name
Number	125000
Date	15/09/2026
Percentage	75%
Formula	=B2*C2

CELL REFERENCES	
Relative	A1
Absolute	\$A\$1
Mixed	A\$1 or \$A1

Figure 2. Excel data types and cell-reference concepts.

Rows, Columns and Ranges

Rows run horizontally and are numbered. Columns run vertically and are lettered. A range is a group of cells, such as A1:C10.

Common Data Types

- Text - names and labels
- Numbers - quantities and amounts
- Dates and times
- Percentages
- Formulas - expressions beginning with =

Good Data Entry

Keep one type of information per column and use clear headings. Avoid mixing unrelated data in the same cell.

Step-by-Step Practical

1. Enter headings Name, Course, Fee and Status in A1:D1.
2. Enter five sample learner records.
3. Select the range A1:D6.
4. Rename Sheet1 to Learners.

Review Questions

1. What is a cell address?
2. What is a range?
3. Name four common data types.
4. Why should columns have clear headings?

Assignment

Create a 10-record student data table.

LESSON 3: ENTERING, EDITING AND MANAGING DATA

Accurate data entry is essential because formulas and reports depend on the values stored in the worksheet.

Cells, Data Types and References

Every value belongs in a cell address such as B4

COMMON DATA TYPES	
Text	Student Name
Number	125000
Date	15/09/2026
Percentage	75%
Formula	=B2*C2

CELL REFERENCES	
Relative	A1
Absolute	\$A\$1
Mixed	A\$1 or \$A1

Figure 2. Data types and references used during data entry.

Editing Cells

Double-click a cell or use the Formula Bar to edit its contents. Delete clears selected content; Undo reverses many recent actions.

AutoFill and Fill Handle

The fill handle can copy values, formulas and recognized sequences such as months or numbers.

Find and Replace

Find locates specific content; Replace can change repeated content efficiently. Always review replacements before applying them widely.

Insert and Delete

Rows, columns and cells can be inserted or deleted. Be careful when formulas depend on the affected range.

Step-by-Step Practical

1. Create a list of January to June using AutoFill.
2. Enter numbers 1 and 2, select both and drag the fill handle to continue the sequence.
3. Use Find to locate one learner name.
4. Correct a misspelled course name.

Review Questions

1. What is AutoFill?

2. What is the fill handle?
3. When is Find and Replace useful?
4. Why should you be careful when deleting rows?

Assignment

Prepare a clean 20-row attendance register.

LESSON 4: FORMATTING PROFESSIONAL WORKSHEETS

Formatting improves readability and helps users understand headings, totals, dates, currency and important results.

Professional Worksheet Formatting

Use formatting to communicate hierarchy

POOR FORMAT

name	score	status
ada	78	pass
tunde	91	pass
mira	64	pass

IMPROVED FORMAT

Student	Score	Status
Ada	78	Pass
Tunde	91	Pass
Mira	64	Pass

Figure 5. Before-and-after worksheet formatting.

Font and Alignment

Use readable fonts, consistent sizes and deliberate alignment. Headings should be visually distinct from data.

Number Formats

- General
- Number
- Currency or Accounting
- Percentage
- Date
- Time

Borders and Fill

Borders define tables while fill colours can identify headings or categories. Avoid excessive decoration.

Column Width and Row Height

Use AutoFit or adjust dimensions so content is readable without unnecessary empty space.

Step-by-Step Practical

1. Format a heading row in bold.
2. Apply currency format to a Fee column.
3. Apply percentage format to a Completion column.

4. AutoFit all used columns.
5. Add a clear border to the table.

Review Questions

1. Why is number formatting important?
2. What is AutoFit?
3. When should percentage format be used?
4. What makes a worksheet look professional?

Assignment

Reformat your attendance register into a professional table.

LESSON 5: FORMULAS AND OPERATORS

A formula is an instruction Excel calculates. Formulas normally begin with the equals sign (=).

Worked Example - Monthly Training Budget

Use formulas instead of typing calculated totals manually

Item	Qty	Unit Cost	Amount
Internet	1	25,000	25,000
Printing	4	5,000	20,000
Transport	8	3,500	28,000
Materials	10	2,000	20,000
TOTAL			93,000

Amount formula example: `=B2*C2` Total formula: `=SUM(D2:D5)`

Figure 4. A worked budget example using formulas.

Arithmetic Operators

- + addition
- - subtraction
- * multiplication
- / division
- ^ exponentiation

Order of Operations

Excel follows mathematical precedence. Parentheses can make the intended calculation clearer.

Using Cell References

Instead of typing values repeatedly, refer to cells. For example, `=B2*C2` calculates quantity multiplied by unit cost.

Formula Safety

Check formulas after copying them. A correct-looking result can still come from an incorrect range.

Step-by-Step Practical

1. Create columns Item, Quantity, Unit Cost and Amount.
2. Enter five expense items.
3. In D2 type `=B2*C2`.
4. Copy the formula down to the remaining rows.

5. Change one quantity and observe the recalculation.

Review Questions

1. What character begins a formula?
2. What does * mean in Excel?
3. Why use cell references?
4. What happens when an input value changes?

Assignment

Create a five-item office expense sheet using formulas.

LESSON 6: CORE FUNCTIONS: SUM, AVERAGE, MAX, MIN AND COUNT

Functions are predefined formulas that perform common calculations.

Formulas and Functions

A formula begins with = and can use cell references

Total	=SUM(B2:B6)
Average	=AVERAGE(B2:B6)
Highest	=MAX(B2:B6)
Lowest	=MIN(B2:B6)
Count	=COUNT(B2:B6)

Figure 3. Core Excel formulas and functions.

SUM

Adds numbers. Example: =SUM(D2:D10).

AVERAGE

Returns the arithmetic mean. Example: =AVERAGE(B2:B10).

MAX and MIN

MAX returns the largest value; MIN returns the smallest.

COUNT

Counts cells containing numbers. COUNTA can count non-empty cells.

Function Arguments

The values or ranges inside parentheses are arguments. Always verify that the selected range contains the intended data.

Step-by-Step Practical

1. Enter ten student scores.
2. Calculate the total with SUM.
3. Calculate the average with AVERAGE.

4. Find the highest score with MAX and lowest with MIN.
5. Count the numeric scores with COUNT.

Review Questions

1. What does SUM do?
2. What does AVERAGE return?
3. Differentiate MAX and MIN.
4. What does COUNT count?

Assignment

Build a score summary showing total, average, highest, lowest and count.

LESSON 7: CELL REFERENCES AND COPYING FORMULAS

Cell references control how formulas behave when copied to other cells.

Cells, Data Types and References

Every value belongs in a cell address such as B4

COMMON DATA TYPES	
Text	Student Name
Number	125000
Date	15/09/2026
Percentage	75%
Formula	=B2*C2

CELL REFERENCES	
Relative	A1
Absolute	\$A\$1
Mixed	A\$1 or \$A1

Figure 2. Relative, absolute and mixed cell references.

Relative Reference

A1 changes relative to the new position when a formula is copied.

Absolute Reference

\$A\$1 remains fixed when copied. This is useful for a tax rate, exchange rate or other constant stored in one cell.

Mixed Reference

A\$1 fixes the row; \$A1 fixes the column.

Practical Example

If B2 contains an amount and F1 contains a fixed rate, =B2*\$F\$1 keeps the rate cell fixed when copied downward.

Step-by-Step Practical

1. Enter five amounts in B2:B6.
2. Enter 7.5% in F1.
3. In C2 type =B2*\$F\$1.
4. Copy C2 down to C6.
5. Check that every copied formula still points to \$F\$1.

Review Questions

1. What is a relative reference?
2. What is an absolute reference?
3. What does \$A1 fix?
4. Why is \$F\$1 useful for a constant rate?

Assignment

Create a price table using one fixed discount-rate cell.

LESSON 8: LOGICAL FUNCTIONS AND IF

Logical functions allow a worksheet to make decisions based on conditions.

Logical Function - IF

Test a condition and return one result or another

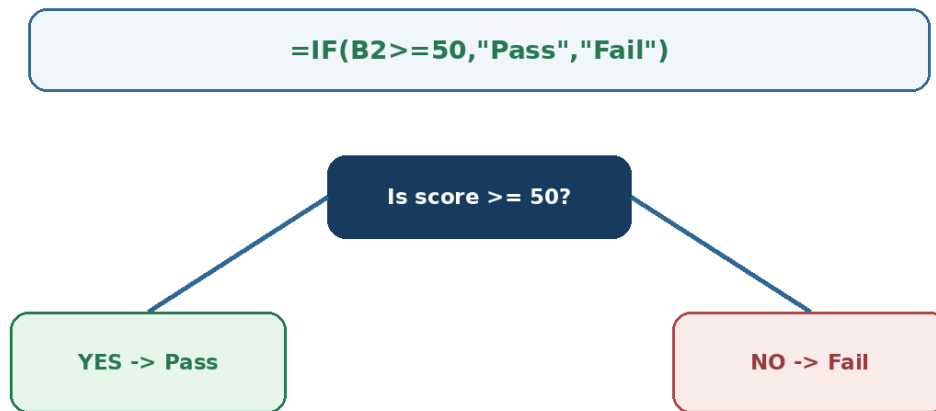


Figure 7. How an IF formula evaluates a condition.

IF Function

The IF function tests a condition and returns one result when TRUE and another when FALSE. Example: `=IF(B2>=50,"Pass","Fail")`.

Comparison Operators

- > greater than
- < less than
- >= greater than or equal to
- <= less than or equal to
- = equal to
- <> not equal to

AND and OR

AND requires all specified conditions to be true. OR requires at least one specified condition to be true.

Nested Logic

Complex formulas can become difficult to audit. Keep business rules clear and test several cases.

Step-by-Step Practical

1. Enter five scores.
2. Use IF to return Pass for scores of 50 or above and Fail otherwise.
3. Test the formula with scores below, equal to and above 50.

4. Add a second condition for a simple eligibility example using AND.

Review Questions

1. What does IF do?
2. What does $\geq$ mean?
3. Differentiate AND and OR.
4. Why should logical formulas be tested with different cases?

Assignment

Create a simple result sheet that automatically assigns Pass or Fail.

LESSON 9: SORTING, FILTERING AND EXCEL TABLES

Sorting changes the order of records. Filtering temporarily displays only records that match selected criteria.

Sorting and Filtering

Sort changes order; Filter temporarily shows matching records

SORT		FILTER	
Before:		Show only Status = Active	
Ada	78	Ada	Active
Tunde	91	Tunde	Active
Mira	64	Hidden temporarily:	
After: Highest to Lowest		Mira	Completed
Tunde	91		
Ada	78		
Mira	64		

Figure 6. The difference between sorting and filtering.

Sorting

Sort text alphabetically, numbers smallest-to-largest or largest-to-smallest, and dates oldest-to-newest or newest-to-oldest.

Filtering

Filters help focus on selected categories without deleting other records.

Excel Tables

Converting a range to a table provides structured formatting, filter controls and easier expansion.

Data Integrity

Select the whole dataset before sorting so related values remain on the same row.

Step-by-Step Practical

1. Create a 10-record course list.
2. Sort by Fee from highest to lowest.
3. Filter Status to show Active only.
4. Convert the range to an Excel Table.
5. Clear the filter and confirm all records return.

Review Questions

1. What is sorting?
2. What is filtering?
3. Why should the entire dataset be included in a sort?
4. Give two benefits of Excel Tables.

Assignment

Prepare and filter a learner enrolment table.

LESSON 10: DATA VALIDATION AND CONDITIONAL FORMATTING

Data validation controls what users can enter, while conditional formatting changes appearance when rules are met.

Data Validation and Conditional Formatting

Control input and make important values visible

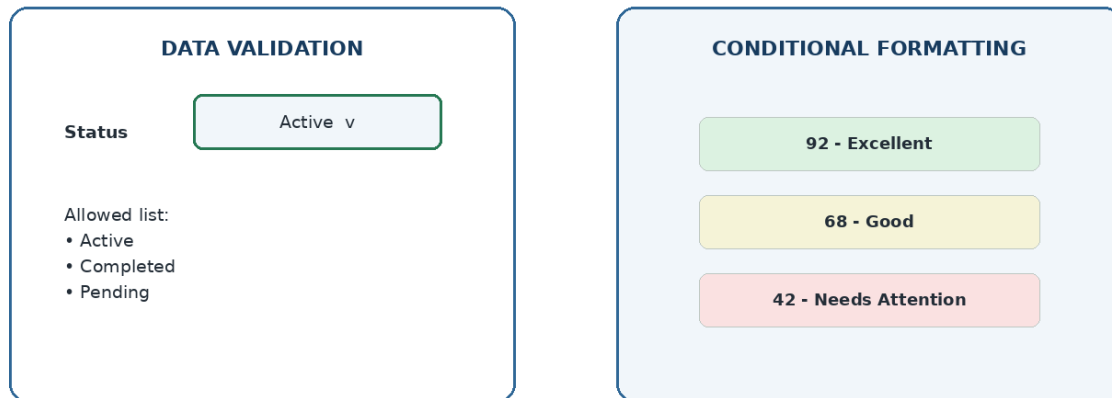


Figure 11. Validation controls and conditional formatting.

Validation Lists

A drop-down list can restrict entries to approved values such as Active, Pending and Completed.

Validation Rules

Rules can restrict numbers, dates, text length or other input.

Conditional Formatting

Rules can highlight high scores, overdue dates, duplicates or values outside an expected range.

Use with Care

Formatting should support decisions. Too many colours and rules can make a sheet harder to understand.

Step-by-Step Practical

1. Create a Status column.
2. Apply a validation list: Active, Pending, Completed.
3. Apply conditional formatting to highlight scores below 50.
4. Enter several test values and confirm the rules behave correctly.

Review Questions

1. What is data validation?

2. What is a drop-down list?
3. What is conditional formatting?
4. Give two useful conditional-formatting examples.

Assignment

Create a controlled course-registration sheet.

LESSON 11: CHARTS AND DATA VISUALIZATION

Charts convert data into visual form so patterns and comparisons can be understood quickly.

Charts and Data Visualization

Choose a chart that matches the question

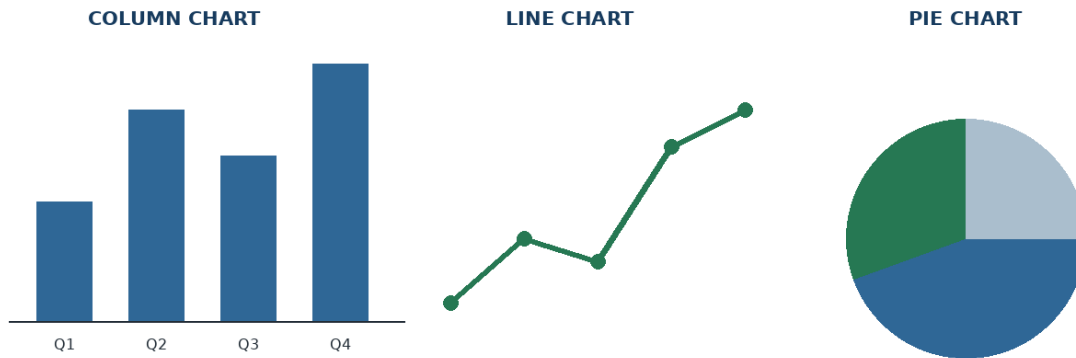


Figure 8. Column, line and pie chart examples.

Column or Bar Charts

Useful for comparing categories.

Line Charts

Useful for showing trends over time.

Pie Charts

Useful for showing parts of a whole when there are only a few categories.

Chart Elements

- Chart title
- Axes where applicable
- Legend where useful
- Data labels when they improve interpretation

Good Chart Practice

Choose a chart based on the question. Avoid 3D effects or decoration that makes values harder to compare.

Step-by-Step Practical

1. Create a monthly enrolment table.
2. Insert a column chart.
3. Add a clear chart title.

4. Create a line chart from monthly revenue.
5. Compare which chart communicates each question better.

Review Questions

1. When is a line chart useful?
2. What is a column chart good for?
3. When should pie charts be used carefully?
4. Name three chart elements.

Assignment

Create two charts from the same dataset and explain which is more useful.

LESSON 12: TEXT, DATE AND USEFUL BUSINESS FUNCTIONS

Excel includes functions for cleaning text, working with dates and preparing business data.

Formulas and Functions

A formula begins with = and can use cell references

Total	=SUM(B2:B6)
Average	=AVERAGE(B2:B6)
Highest	=MAX(B2:B6)
Lowest	=MIN(B2:B6)
Count	=COUNT(B2:B6)

Figure 3. Functions are reusable formulas for common tasks.

Text Functions

- LEFT extracts characters from the beginning
- RIGHT extracts characters from the end
- LEN counts characters
- TRIM removes unnecessary spaces
- CONCAT or & joins text

Date Functions

- TODAY returns the current date
- YEAR extracts a year from a date
- MONTH extracts a month number
- DAY extracts a day number

Rounding

ROUND can control the number of decimal places in calculated results.

Practical Relevance

These functions help clean imported data, build identifiers and calculate time-based reports.

Step-by-Step Practical

1. Enter five full names with accidental extra spaces and clean them with TRIM.
2. Extract the first three characters of a course code with LEFT.

3. Enter dates and extract YEAR and MONTH.
4. Round a calculated amount to two decimal places.

Review Questions

1. What does TRIM do?
2. What does LEN return?
3. What does TODAY return?
4. Why is ROUND useful?

Assignment

Create a small data-cleaning worksheet using at least four functions.

LESSON 13: LOOKUP FUNCTIONS

Lookup functions search for a key and return related information from a table.

Lookup Concept

Find a key in a table and return related information

Course Code

EDA-EXC-103

Code	Course	Fee
EDA-WRD-102	Microsoft Word	55,000
EDA-EXC-103	Microsoft Excel	65,000
EDA-PPT-104	PowerPoint	55,000

Result:
Microsoft Excel
65,000

Figure 9. Looking up a course code in a reference table.

Lookup Idea

A course code can be used to retrieve a course name or fee from a reference table.

XLOOKUP

In supported Excel versions, XLOOKUP provides a flexible way to search one range and return a value from another.

VLOOKUP

VLOOKUP is widely used in older workbooks. It searches the first column of a table and returns a value from a specified column.

Exact Matches

For codes, IDs and other identifiers, exact-match logic is usually appropriate.

Error Handling

If a key is missing, the formula may return an error. Validate the key and consider appropriate error handling.

Step-by-Step Practical

1. Create a reference table with Course Code, Course and Fee.
2. Enter one course code in a separate cell.
3. Use an available lookup function to return the course name.
4. Return the course fee.

5. Test a code that does not exist and observe the result.

Review Questions

1. What is a lookup?
2. What does a lookup key represent?
3. Why are exact matches useful for IDs?
4. What should you check when a lookup fails?

Assignment

Create a course-fee lookup tool.

LESSON 14: PIVOTTABLES AND PIVOTCHARTS

A PivotTable summarises large datasets by categories without changing the original source records.

PivotTable Concept

Summarise many rows without rewriting the source data

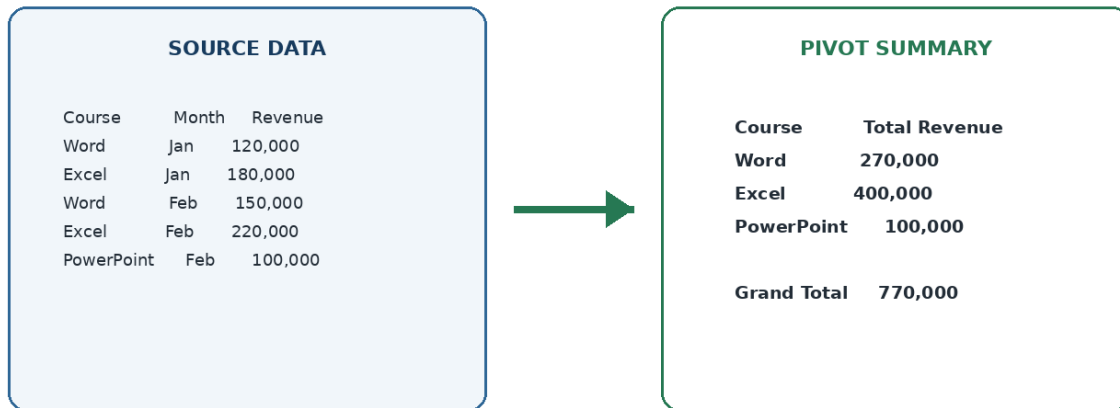


Figure 10. From detailed source data to a PivotTable summary.

Source Data

Good PivotTables begin with clean tabular data: one header row, consistent columns and no unnecessary blank rows.

Rows, Columns, Values and Filters

Fields can be arranged to group categories and calculate totals, counts or averages.

Refreshing

If source data changes, the PivotTable may need to be refreshed.

PivotCharts

A PivotChart provides a visual summary connected to the PivotTable.

Step-by-Step Practical

1. Create sample sales or enrolment data with Course, Month and Revenue.
2. Insert a PivotTable.
3. Place Course in Rows and Revenue in Values.
4. Add Month as a filter or column.
5. Create a PivotChart and refresh after changing source data.

Review Questions

1. What is a PivotTable?
2. Why must source data be clean?
3. What does the Values area contain?
4. When should a PivotTable be refreshed?

Assignment

Build a PivotTable summarising revenue by course.

LESSON 15: PAGE LAYOUT, PRINTING AND PDF EXPORT

A correct worksheet can still print badly if page layout is not prepared.

Worked Example - Monthly Training Budget

Use formulas instead of typing calculated totals manually

Item	Qty	Unit Cost	Amount
Internet	1	25,000	25,000
Printing	4	5,000	20,000
Transport	8	3,500	28,000
Materials	10	2,000	20,000
TOTAL			93,000

Amount formula example: `=B2*C2` Total formula: `=SUM(D2:D5)`

Figure 4. Portrait and landscape layout concepts also apply to spreadsheet reports.

Print Area

Set a print area when only part of a worksheet should be printed.

Orientation and Margins

Portrait suits many narrow reports; landscape may suit wide tables.

Scaling

Fit-to-page options can help, but excessive shrinking makes text unreadable.

Headers and Footers

Use them for report titles, dates, confidentiality labels or page numbers where appropriate.

PDF Export

Review the final PDF to ensure columns, titles and page breaks appear correctly.

Step-by-Step Practical

1. Open Print Preview for a wide worksheet.
2. Compare portrait and landscape orientation.
3. Adjust margins and scaling.
4. Repeat the heading row on printed pages if your version supports it.
5. Export the worksheet to PDF and inspect the PDF.

Review Questions

1. What is Print Area?
2. When is landscape orientation useful?
3. Why can excessive scaling be a problem?
4. What should you check after PDF export?

Assignment

Prepare a two-page professional report for printing and PDF.

LESSON 16: DATA CLEANING AND ERROR CHECKING

Reliable analysis depends on clean, consistent data and correctly constructed formulas.

Data Validation and Conditional Formatting

Control input and make important values visible

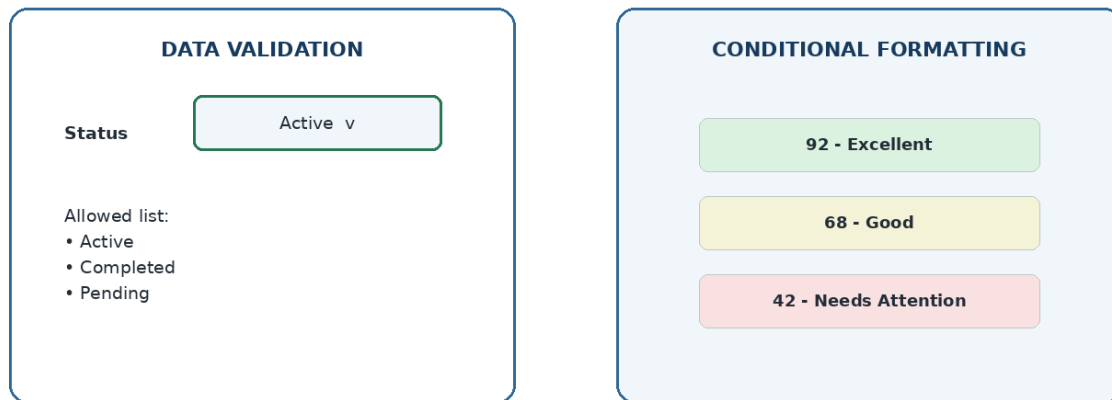


Figure 11. Controlled inputs and visual checks improve data quality.

Common Data Problems

- Duplicate records
- Leading or trailing spaces
- Numbers stored as text
- Inconsistent date formats
- Blank required fields
- Misspelled categories

Formula Errors

Common Excel errors can indicate invalid references, unavailable values or inappropriate operations. Investigate the cause instead of hiding the error blindly.

Remove Duplicates

Use carefully and keep a backup when the dataset is important.

Quality Checks

Compare totals, inspect unusual values and test formulas at the beginning, middle and end of a copied range.

Step-by-Step Practical

1. Create a deliberately messy dataset.
2. Use TRIM or other tools to clean text.
3. Identify duplicates.

4. Check for numbers stored as text.
5. Audit copied formulas in several rows.

Review Questions

1. Why does data cleaning matter?
2. Give four common data problems.
3. Why should duplicates be removed carefully?
4. How can you audit copied formulas?

Assignment

Clean and document a 20-row practice dataset.

LESSON 17: BUSINESS REPORTING AND DASHBOARD DESIGN

A dashboard brings important indicators and visuals together so decision-makers can understand performance quickly.

Excel Dashboard Example

Combine KPIs, tables and charts into one decision view

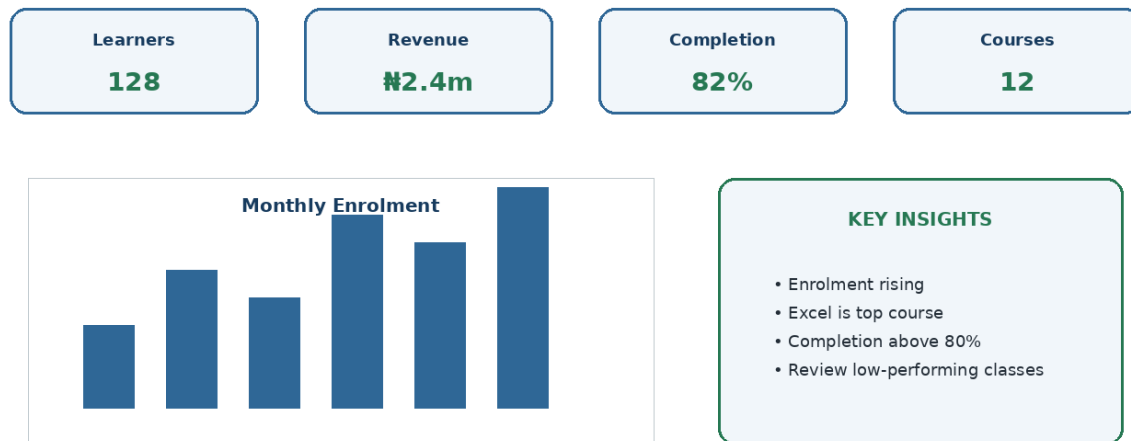


Figure 12. A one-page management dashboard example.

Key Performance Indicators

KPIs are selected measures such as enrolment, revenue, completion rate or average score.

Dashboard Structure

A useful dashboard may contain KPI cards, a summary table, charts, filters and a short insight section.

Consistency

Use consistent number formats, titles and periods. A dashboard should answer specific questions rather than display every available metric.

Decision Focus

The purpose is not decoration. Each visual should support a decision or explain performance.

Step-by-Step Practical

1. Create a summary sheet.
2. Calculate four KPIs.
3. Create one trend chart and one comparison chart.
4. Arrange the elements into a one-page dashboard.
5. Write three insights from the dashboard.

Review Questions

1. What is a KPI?
2. What is the purpose of a dashboard?
3. Why should dashboard metrics use consistent periods?
4. What makes a dashboard decision-focused?

Assignment

Create an Ethan Digital Academy training-performance dashboard.

LESSON 18: FINAL PROFESSIONAL EXCEL PROJECT

The final project combines data entry, formulas, functions, analysis, visualization and reporting.

Excel Dashboard Example

Combine KPIs, tables and charts into one decision view

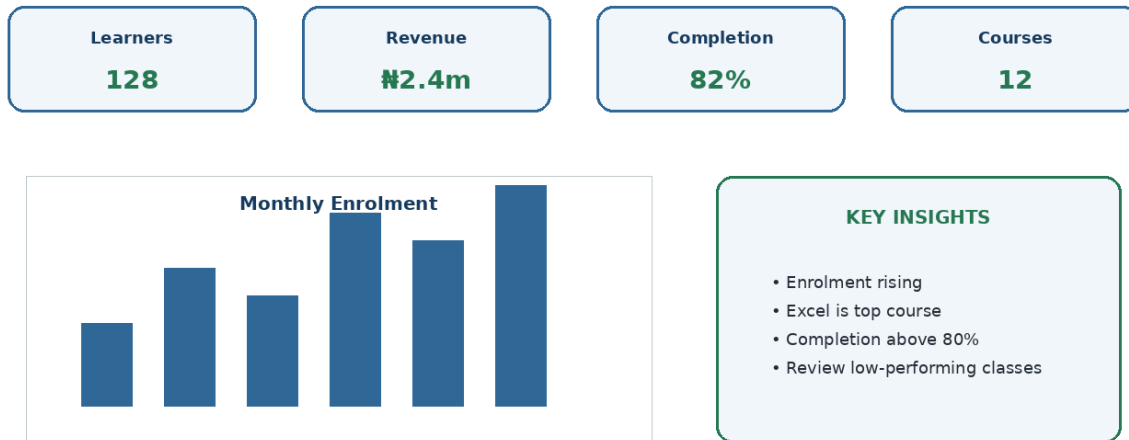


Figure 12. The final workbook should lead to a clear management dashboard.

Project Scenario

You are preparing an Excel management workbook for Ethan Digital Academy.

Required Sheets

- Learners - learner records and status
- Payments - course fees and payment records
- Attendance - attendance percentages
- Results - assessment scores and Pass/Fail logic
- Dashboard - KPIs and charts

Required Skills

- Professional formatting
- SUM, AVERAGE, MAX, MIN and COUNT
- Relative and absolute references
- IF logic
- Sorting and filtering
- Data validation
- Conditional formatting
- Lookup function
- PivotTable
- Charts
- Print/PDF preparation

Quality Standard

The workbook must be accurate, readable, logically organized and easy for another user to understand.

Step-by-Step Practical

1. Build the five required worksheets.
2. Enter realistic sample data.
3. Add formulas and functions.
4. Create at least one lookup and one PivotTable.
5. Build the dashboard.
6. Check formulas and formatting.
7. Prepare the Dashboard sheet for PDF export.

Review Questions

1. Which skills does the project combine?
2. Why should a workbook use separate logical sheets?
3. What quality checks should be completed before submission?
4. How does the dashboard help management?

Assignment

Submit the complete professional workbook and exported dashboard PDF.

REAL-LIFE CASE STUDY - ETHAN DIGITAL ACADEMY

Ethan Digital Academy needs a workbook that tracks learners, payments, attendance and results. The administrator creates structured tables, uses formulas to calculate totals and balances, applies IF logic to classify outcomes, validates status entries, creates a course-fee lookup, summarizes revenue with a PivotTable and presents key indicators on a dashboard. The result is a workbook that supports both daily operations and management decisions.

FINAL PRACTICAL PROJECT - 100 MARKS

1. Create the Learners worksheet with at least 20 records.
2. Create a Payments worksheet and calculate amounts, balances and totals.
3. Create an Attendance worksheet and calculate attendance percentages.
4. Create a Results worksheet using formulas and IF logic.
5. Create a reference table and a working course lookup.
6. Apply data validation and conditional formatting.
7. Create a PivotTable summary.
8. Create at least two appropriate charts.
9. Create a one-page Dashboard sheet with four KPIs.
10. Prepare the dashboard for PDF export and submit the workbook.

QUIZ - 30 QUESTIONS

1. What is a workbook?
2. What is a worksheet?
3. What is a cell address?
4. What is a range?
5. Name four Excel data types.
6. What character starts a formula?
7. What does SUM do?
8. What does AVERAGE do?
9. What does MAX return?
10. What does MIN return?
11. What does COUNT do?
12. What is a relative reference?

13. What is an absolute reference?
14. What does \$A\$1 mean?
15. What is the purpose of IF?
16. What does >= mean?
17. Differentiate AND and OR.
18. What is sorting?
19. What is filtering?
20. What is an Excel Table?
21. What is data validation?
22. What is conditional formatting?
23. When is a line chart useful?
24. When is a column chart useful?
25. What does TRIM do?
26. What is a lookup function used for?
27. What is a PivotTable?
28. Why should PivotTables be refreshed?
29. What is a KPI?
30. Why should a workbook be checked before PDF export?

FINAL EXAMINATION - 100 MARKS

Section A - Objective Questions (20 Marks)

Twenty questions covering Excel terminology, formulas, functions and data-management concepts.

Section B - Short Answer and Formula Questions (30 Marks)

Explain spreadsheet concepts and write appropriate formulas for realistic scenarios.

Section C - Practical Assessment (50 Marks)

Build and format a workbook, use formulas/functions, apply validation and conditional formatting, analyse data with a PivotTable, create charts and prepare a professional dashboard.

COURSE COMPLETION OUTCOME

Graduates of EDA-EXC-103 Microsoft Excel will be able to build, calculate, analyse and present professional spreadsheets for academic, office and business use. They will be prepared for advanced Excel Data Analysis, Power BI, Business Intelligence and reporting courses.